

Vincent Zhou

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EDUCATION

University of Waterloo

Sept 2025 – June 2030

Bachelor of Applied Science, Computer Engineering, Honours Co-op

EXPERIENCE

Waterloo Aerial Robotics Group (WARG)

Autonomy Project Manager

Sept 2026 – Present

- Manage a **30-member** autonomy subteam, organizing contributors into parallel project groups, scoping the flight-software backlog into issues sized for onboarding, and tracking them through to flight readiness for the 2027 AEAC competition.
- Run code review for the autonomy stack and set its architectural direction, defining the ROS 2 node and message boundaries between the perception, GNC, and obstacle-avoidance packages so work split across groups integrates without rework.

Software Engineering Intern

May 2026 – Sept 2026

- Independently designed, built, and flight-tested a **visual teach-and-repeat** precision-landing system (**ROS 2**, **OpenCV**, **MAVROS**) that records an altitude-keyed map of **ORB** features during takeoff, then on descent matches live frames against the nearest teach frame using Lowe-ratio-filtered kNN matching, metric back-projection with roll/pitch compensation, and **RANSAC** affine estimation to guide the drone back onto its launch point, landing within **15 cm in flight test** and **2 cm in simulation**.
- Validated the descent controller in isolation against an **AprilTag fiducial baseline** first, confirming landings on the marker before substituting feature-based localization, so control and perception failures could be diagnosed separately.
- Hardened the pipeline against field failure modes: an altitude-proportional alignment cone that tapers descent authority with match quality, a stale-vision recovery state machine that climbs to widen the camera footprint and re-acquire, teach-frame keypoint gating, and inlier-ratio and scale validation to reject degenerate transforms.
- Architected the non-optimal teach-and-repeat pipeline on an **NVIDIA Jetson**: a custom behaviour tree orchestrating the full mission, a **PI** controller streaming body-frame velocity setpoints through **MAVROS**, and a **CUDA**-accelerated brute-force ORB matcher running at **10 Hz**, with final touchdown handed to **ArduPilot** LAND under debounced ground confirmation.
- Architected a full-stack, real-time **UAV geolocation platform** (**React/TypeScript** frontend, **Flask** REST API) for the 2026 AEAC Firefighting Competition, enabling ground operators to capture, annotate, and geolocate targets spotted mid-flight using a pinhole camera model and ray-plane intersection against live telemetry to localize targets in 3D to **<0.5 m**.
- Built TCP client/server capture pipelines and UDP MAVLink attitude/rangefinder streams for real-time UAV image and telemetry sync.

PROJECTS

SITL-Plus | UAV Software-in-the-Loop Simulator, WARG

- Built a from-scratch simulator using **PyBullet** rigid-body physics to model a quadcopter's flight dynamics, bridged to **ArduPilot**'s SITL binary over a **UDP** JSON protocol for high-rate physics/flight-controller state exchange.
- Simulated onboard sensors: downward and forward RGB-D cameras via **PyBullet**'s **OpenGL** renderer and a ray-cast rangefinder/LiDAR, streamed to **Rerun** over **gRPC** for live 3D visualization, enabling integration testing of the vision and precision-landing pipelines without physical hardware.
- Containerized the stack (physics engine + **ArduPilot** SITL/MAVProxy) with **Docker Compose**, covered the sensor and dynamics models with a **pytest** suite, and used the simulator to validate and tune the precision-landing pipeline to **2 cm** before committing to flight tests.

Classly | Classroom Management Platform: React 19, Node.js/Express 5, TypeScript

- Built a full-stack classroom platform (classes, invites, enrollments, weighted grades) on **Postgres/Neon + Drizzle**, with role-based access (admin/teacher/student) and **Better Auth** sessions (email/password + Google/GitHub OAuth) behind a Vercel api proxy for first-party cookies.
- Migrated the API from a long-running **Render** service onto **Cloudflare Workers**, cutting idle cold starts from 10–30s to near-instant; kept Express via Node compatibility, used Neon HTTP for reads/writes and request-scoped WebSocket pools for transactions, and added cron-based session cleanup with secrets managed via **Wrangler**.
- Hardened concurrent enrollment/grade writes with pessimistic transactional locking, and secured the API with **Arcjet** bot detection/rate limiting and parameterized queries.

MacroMe | Autonomous web agent

2nd Place, WebAgent Track

Battle of the Schools (UTMIST × Wat.ai)

- Built an autonomous DoorDash web agent in **TypeScript/Node.js** using **Steel** cloud browsers and **Playwright** (CDP) with a persisted login profile to discover nearby restaurants, scrape menus, and assemble carts; used an **LLM** (**OpenRouter**) to rank meals against per-slot macro and budget targets, with checkout recovery, live session streaming, and human approval before Place order.
- Integrated **SQLite FTS5** retrieval over USDA nutrition data to inform meal ranking, and built a **custom MCP server** exposing daily macro summaries, saved digests, and email delivery to AI assistants.

Neural Network From Scratch | C++, Eigen

- Implemented a multilayer perceptron from scratch with no ML framework, deriving forward and backpropagation by hand, with mini-batch SGD and early stopping that checkpoints the best weights on validation plateaus, reaching **95.4%** test accuracy on MNIST.

SKILLS

- **Languages:** Python, C++, TypeScript/JavaScript, Java, SQL, Bash, Lua, Verilog
- **Robotics:** ROS 2, ArduPilot/MAVROS, MAVLink, OpenCV, CUDA, PyBullet, Rerun, NumPy, Eigen, NVIDIA Jetson, Raspberry Pi
- **Backend & Agents:** FastAPI, Flask, Node.js/Express, REST, PostgreSQL, Drizzle ORM, Neon, Pydantic, Zod, Playwright, CDP, Steel, MCP, OpenRouter, PyTorch, TCP/UDP
- **Infrastructure & Testing:** Docker, Cloudflare Workers/Pages, Wrangler, Git/GitHub, GitHub Actions CI, Linux, pytest, Vitest, Quartus/FPGA